MONEY MATH, ONE PAGE

Compound interest

Interest that earns interest. One formula covers savings accounts, loans and index funds. Everything else on this page is a rearrangement of it.

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

- A what you end up with
- P what you put in
- r annual rate, as a decimal
- n compounding periods per year
- t years

Compounding all the time

As $n \to \infty$, the bracket converges to e^r for each year:

$$A = Pe^{rt}$$

The rule of 72

Doubling time, solved from $2P = P(1+r)^t$:

$$t_{2\times} = \frac{\ln 2}{\ln(1+r)} \approx \frac{72}{100\,r}$$

Monthly contributions

Add c every month at monthly rate $i = r/12$, for m months:

$$FV = c \cdot \frac{(1+i)^m - 1}{i}$$

The rate you actually get

The annual percentage yield folds compounding into one number:

$$\mathrm{APY} = \left(1 + \tfrac{r}{n}\right)^n - 1$$

Worked example

Put $10,000 in at 5% compounded monthly for 30 years:

$$A = 10{,}000\left(1 + \frac{0.05}{12}\right)^{12\cdot 30} \approx \$44{,}677$$

The same money at simple interest grows to only $10{,}000\,(1 + 0.05 \cdot 30) = \$25{,}000$. The gap of about $19,677 is interest on interest.